

Stereotactic Photon Beam Irradiation of Uveal Melanoma: Indications and Experience at the University of Vienna since 1997

Karin Dieckmann¹, Dietmar Georg¹, Martin Zehetmayer², Andrea Rottenfusser¹, Richard Pötter¹

Introduction

Most uveal melanomas are currently treated by eye-preserving radiotherapy (1, 2, 3). No advantage of enucleation of posterior uveal melanoma (< 12 mm) could be demonstrated in multiple studies (e.g. [4]). Besides brachytherapy and highly conformal proton therapy advanced photon beam therapy techniques have been introduced during the last years [5]. Since 1997, at the Medical University of Vienna, patients with uveal melanoma have been treated with 3D stereotactic photon beam radiotherapy. Special fixation- and eye-monitoring systems have been developed to perform a local treatment which provides high local tumor control with acceptable long term side effects [6, 7].

Material and Methods

More than 158 patients with uveal melanoma were included in our treatment protocol with either tumors thicker being than 7 mm or with posterior pole tumors smaller in thickness (> 2.5 mm) but a central margin within 3 mm to optic disc rim or macula, see table 1. At time of diagnosis, full ophthalmologic examination was performed with ultrasonographic measurements of basal tumor dimensions, tumor thickness, and estimation of tumor to disk and tumor to macula distance. The size, shape and location of the tumors were further defined by means of high resolution CT and MRI during tumor delineation for treatment planning. All patients were examined with hepatic ultrasound, imaging, and dermatologic examinations.

Treatment planning: Treatment planning and treatment performance of uveal melanoma at a linac were based on a commercial thermoplastic stereotactic head mask (BrainLAB, Heimstetten, Germany), which was modified to enable eye fixation and eye-monitoring [6]. The used fixation system is CT and MRI compatible and the software provides a comparison of treatment preparation and treatment delivery information. During treatment planning and delivery all patients were instructed to fixate a light source and a mini-video camera was used to monitor the eye and pupil position. Stored pre-treatment and on-line treatment images of the eye, provided by the mini-video camera, were compared and treatment was interrupted if necessary [8].

The control system gives a trigger signal online to the linac to interrupt automatically irradiation in case of deviation of the actual pupil position compared with the planning pupil position. The pupil position is verified approximately 15 times per second.

For treatment planning a computerized 3-D model of the eye and tumor is generated prior to treatment. High resolution MR- and CT images are fused, so that the images can be viewed in axial, sagittal and coronal planes. The PTV was defined as the contrast medium enhanced tumor visible on MRI (CTV) plus a safety margin of about 2 mm margin in all directions. Depending on the tumor shape, arc beam SRT with four to seven arcs per isocenter or static conformal SRT with 8 to 12 beams were applied (Figure 1). All patients were treated with 6 MV at a linac

Table 1. Patients' characteristics.

Total number of patients	n = 158	
Involved eye	Left	81
	Right	77
Gender	Male	92
	Female	66
Tumor shape	Dome	128
	Mushroom	30
Age	59.5 ± 13.9 years	(range 21–89)
Visual acuity	0.516 ± 0.34	
Acuity better than 0.1	139	
Initial tumor volume	329 ± 261 mm ³	(range 34–1950 mm ³)
Tumor thickness	≤ 3 mm	n = 11
	> 3 mm	n = 147
Distance tumor macula	0 mm	n = 51
	> 0 ≤ 3 mm	n = 93
	> 3 mm	n = 14
Distance to optic nerve	0 mm	n = 40
	> 0 ≤ 3 mm	n = 87
	> 3 mm	n = 31

Key Words: Uveal melanoma · Stereotactic radiotherapy · Conformal radiotherapy

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Departments of ¹Radiotherapy and ²Ophthalmology, University of Vienna, Vienna, Austria

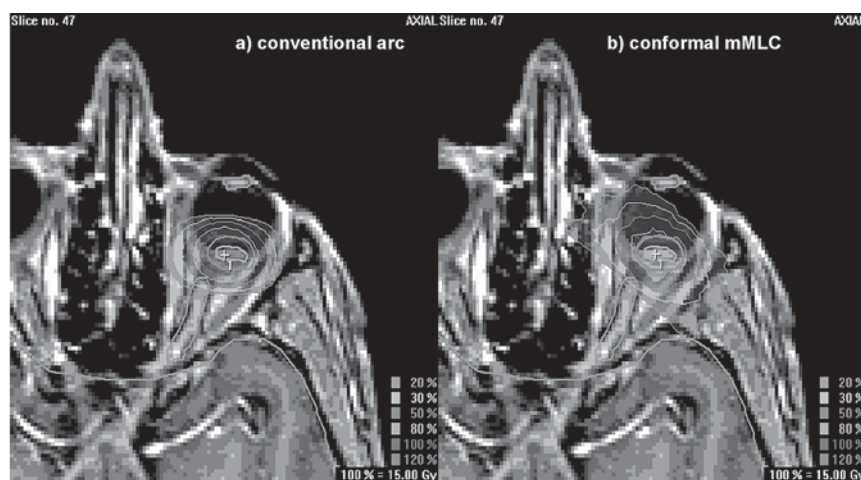


Figure 1. Examples of typical dose distributions for different treatment techniques. a) conventional arc, b) conformal mMLC

with a hypofractionated scheme, either delivering five fractions of 14 Gy or 12 Gy prescribed to the 80% isodose.

Follow-up: Patients were regularly followed by the ophthalmologist 1 month after irradiation, every 3 month for the first 2 years, and every 6 months thereafter. A complete functional assessment, including determination of visual acuity, lid damage, keratitis, cataract, glaucoma, optic neuropathy, maculopathy and ocular inflammation, was performed.

A liver ultrasound was performed every 6 months. Enucleation was performed for local recurrence and for ocular complications such as severe glaucoma, major retinal detachment, disabling eye pain and tumor progression.

Clinical Results

Local control: After a median follow-up period of 33.4 months (range 3 to 85 months), local tumor control, defined as continuous tumor regression or stable disease of the tumor, was achieved in 98% (155/158 patients).

After 1 year, tumor height was reduced or stable in 114 of 130 patients (88%); after 2 years in 83/92 (90%) and after 3 years in 56/62 patients (90%), respectively. The mean reduction of tumor height was 24% at 1 year ($n = 130$), 29% at 2 years ($n = 92$), and 33% at 4 years ($n = 62$). The mean relative tumor volume reductions were 32%, 39%, and 48% at 12, 24, and 48 months, respectively. However, tumor response can vary significantly after radiotherapy. Local recurrences were observed in 2 patients within the first two years.

Secondary enucleation was performed in 23 (14%) patients after a median time of 24 months (range 5 to 87 months). The reasons for enucleation were tumor progression ($n = 2$), neovascular glaucoma with high intraocular pressure and pain, secondary glaucoma, or total retinal detachment in the remaining 21 patients. None of these patients developed distant metastasis or died up to now.

15 (9%) patients developed metastasis after a median observation time of 42 months (range 11 to 77 months). Six patients died within 1 to 16 months after diagnosis of metastasis (median 5.5 months). Seven more patients died due to other reasons, such as age, second cancer, or suicide.

Side effects: Acute side effects in the anterior part of the eye were: blepharo-conjunctivitis (5%), cornea-epithel-defects (3%),

epitheliolysis (5%), and madarosis (6%). They were more common if the tumor is located anterior.

Long term side effects in the posterior part of the eye were: opticopathy in 65 patients (41%) and retinopathy in 70 patients (44%). Median time to opticopathy was 27 months (range 12–72), and median time to retinopathy was 23 months (range 0–60). 23 patients (7%) developed a neovascular glaucoma after a median time of 24 months (range 12–72). This time delay correlated with data in literature on proton beam irradiation where neuropathy and retinopathy were described after 18–30 months [9].

At the time of RT cataract was diagnosed in 31 patients. Due to the radiosensitivity of the lens additionally 30/127 (23%) patients developed a cataract 2 to 4 years after RT. In 19/61 (31%) patients a cataract operation has been performed.

Visual acuity at last examination declined to 0.2 ± 0.29 . Visual acuity of > 0.1 was present only in 60 patients (38%).

Discussion

Stereotactic photon therapy of uveal melanoma, based on CT and MR images, is a safe and precise treatment option. Dose conformation can be as good in SRT for photon technique as for protons (10). As it has been demonstrated that dose reduction to optic nerve or macula is equal for both beam qualities in case where the tumor is at a distance < 2 mm. Only organs at risk, such as the lacrimal glands, were found to profit from proton irradiation [10]. The treatment time is longer for photon than for proton treatment. Due to this fact, the on-line detection of the pupil position and the registration of the deviation are necessary to guarantee a high quality treatment for linac based SRT of uveal melanoma.

Much like proton beam radiotherapy, SRT is useful for tumors that are considered to be excessively large for brachytherapy or because of doubts as to whether a plaque can be positioned accurately over a posterior tumor without causing damage of optic disk or macula. With large tumors, SRT would be contraindicated if the patient does not accept the increased chances of retinal detachment and neovascular glaucoma. Similarly, patients with posterior tumor extension should be treated with this modality only if they accept the increased likelihood of optic neuropathy and maculopathy.

After a median observation time of 33 months, 41% of our analyzed patients developed opticopathy and 44% of the patients retinopathy. It is widely accepted that local side effects are additionally influenced by tumor size, location and the total dose. As described by Flickinger et al, there must be a direct correlation between applied dose, of the optic nerve exposed to high doses and the expected degraded visual acuity and optic nerve atrophy, respectively [11]. Generally visual loss from direct radiation damage to optic nerve and macula is untreatable. However, there is a growing evidence that much of the ocular morbidity following radiotherapy is caused by toxic effects of the residual tumor. Methods for treating these complications are improving thanks to the development of intravitreal steroid injections, transpupillary thermotherapy, and local resection techniques [12, 13, 14, 15].

In particular, stereotactic radiotherapy is contraindicated by tumors diameter greater than 20mm thickness, extensive diffuse

spread, and bulky extraocular extension. Especially in small and medium sized posterior located tumors, SRT offers an elegant non-invasive and patient-friendly approach to control the tumor with comparably mild to acceptable side-effects.

In this study local control is 98% after a median observation time of 33 months. The observation time is still too short to draw definitive conclusions but these results are comparable to the 82–98% local control rate reported by other groups [16, 17]. In less than 2% of our patients tumor re-growth was the reason for enucleation. Therefore the total dose of 60 or 70 Gy for local control was found to be excellent, nevertheless in line with experimental studies, clinical trials with a reduced dose/fractions may be considered in future [18].

Conclusion

Stereotactic photon therapy of uveal melanoma, based on CT and MR images, is a safe and precise treatment option. Local control is excellent with a dose > 60 Gy. The observed side effects were comparable to other treatment options for the specific cohort considered in the present study. Automatic surveillance of the position of the eye allows to perform gated beam delivery for ocular diseases. It would be interesting to initiate a comparative study with protons and photons not only under the aspect of side effects and local control, but as well under the aspect of spiraling health cost in all western countries as mentioned by Bornfeld et al. [19]. Beside the excellent control of SRT with photons of uveal melanomas this method is very cost effective.

References

- Mossbock G, Rauscher T, Winkler P, Kapp KS et al. Impact of Dose rate on Clinical Course in Uveal melanoma after Brachytherapy with Ruthenium-106. *Strahlenther Onkol* 2007;183:571–575.
- Marnitz S, Cordini D, Bendl R, Lemke AJ et al. Proton therapy of uveal melanomas: intercomparison of MRI-based and conventional treatment planning. *Strahlenther Onkol* 2006;182:395–399.
- Hocht S, Stark R, Seiler F, Heufelder J, Bechrakis NE, Cordini D, Marnitz S, Kluge H, Foerster MH, Hinkelbein W. Proton or stereotactic photon irradiation for posterior uveal melanoma? A planning intercomparison. *Strahlenther Onkol*. 2005;181:783–788.
- Ganbrelle J, Grange JD, Devouassoux Shisheboran M, Rivoire M, et al. Survival after primary enucleation for choroidal melanoma: Changes induced by the introduction of conservative therapies. *Graefes Arch Clin Exp Ophthalmol*, 2006;22.
- Zehetmayer M, Kitz K, Menapace R, Ertl A. Suction attachment for stereotactic radiosurgery of intraocular malignancies. *Ophthalmologica* 1994; 208:119–121.
- Bogner J, Petersch B, Georg D, Dieckmann K, Zehetmayer M, Pötter R. A noninvasive eye fixation and computer-aided eye monitoring system for linear accelerator-based stereotactic radiotherapy of uveal melanoma. *Int J Radiat Oncol Biol Phys* 2003;56:1128–1136.
- Dieckmann K, Georg D, Bogner J, Zehetmayer M, Petersch B, Chorvat M, Weitmann H, Pötter R. Optimizing Linac-based stereotactic radiotherapy of uveal melanomas: 7 years' clinical experience. *Int J Radiat Oncol Biol Phys* 2006;66(Suppl.):47–52.
- Petersch B, Bogner J, Dieckmann K, Pötter R, Georg D. Automatic real-time surveillance of eye positioning and gating for stereotactic radiotherapy of uveal melanoma. *Med Phys* 2004;31:3521–3527.
- Höcht S, Bechrakis N, Kreusel KM, Kluge H, Heese J, Heufelder J, et al. Proton therapy of uveal melanomas in Berlin. 5 years of experience at the Hahn-meitner Institute. *Strahlenther Onkol* 2004;180:419–424.
- Weber DC, Bogner J, Verwey J, Georg D, et al. Proton beam radiotherapy versus fractionated stereotactic radiotherapy for uveal melanomas: A comparative study. *Int J Radiat Oncol Biol Phys* 2005;63:373–384.
- Flickinger JC, Konndziolka D, Lunsford LD. Dose and diameter relationships for facial, trigeminal, and acoustic neuropathies following acoustic neuroma radiosurgery. *Radiother Oncol* 1996;41:215–219.
- Damato B, Groenewald C, Mc Galliard J, Wong D. Endoresection of choroidal melanoma. *Br J Ophthalmol* 1998;82:213–218.
- Bechrakis NE, Höcht S, Martus P, Kreusel KM, Heese J, Foerster MH. Endoresektion von grossen uvealen Melanomen. [Endoresection following proton beam irradiation of large uveal melanoma]. *Ophthalmologie* 2004; 101:370–376.
- Shields CL, Demirci H, Dai V, Marr BP, Mashayekhi A, et al. Intravitreal triamcinolone acetonide for radiation maculopathy after plaque radiotherapy for choroidal melanoma. *Retina* 2005;25:868–874.
- Bornfeld N, Talies S, Anastassiou G, Schilling H, et al. Endoresektion maligner Melanome der Uvea nach präoperativer stereotaktischer Einzeldosis-Konvergenzbestrahlung mit dem Leksell-Gamma-knife. [Endoscopic resection of malignant melanomas of the uvea after preoperative stereotactic single dose convergence irradiation with the Resell gamma knife]. *Ophthalmologie* 2002;99:338–344.
- Egger E, Zografos L, Schalenbourg A, et al. Eye retention after proton beam radiotherapy for uveal melanoma. *Int J Radiat Oncol Biol Phys* 2003; 55:867–880.
- Gragoudas ES, Lane AM, Munzenrider J, Egan KM, Li W. Long-term risk of local failure after proton therapy for choroidal/ciliary body melanoma. *Trans Am Ophthalmol Soc* 2002;100:43–8.
- van-den-Aardweg GJ, Kilic E, de-Klein A, Luyten GP. Dose fractionation effects in primary and metastatic human uveal melanoma cell lines. *Invest Ophthalmol Vis Sci* 2003;44:4660–4664.
- Bornfeld N. proton treatment of uveal melanomas. *Strahlenther Onkol* 2007;183:1–2.

Karin Dieckmann, MD

Department of Radiotherapy and Radiobiology
Medical University of Vienna
Waehringer Guertel 18–20
1090 Vienna
Austria
karin.dieckmann@akhwien.at